NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM I

MH2100 – Calculus III

September 2022

TIME ALLOWED: 90 minutes

Name:

Matric. no.:

Tutor group:

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. **ONE** two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1. **(17 marks)**

(i) [3 marks] Find parametric equations for the line that is tangent to the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} t^2 - 2t + 1 \\ \sin(t^2 - 1) \end{pmatrix},$$

at the point $(0, 0)$.

(ii) [3 marks] At the points defined by $t = a$ the tangent line to the curve parametrised by

$$\begin{cases} x(t) = 2t \\ y(t) = 3t^2 + t \\ z(t) = -3t^2 - t \end{cases}$$

contains the point $(0, -12, 12)$. Find all such values of a.

(iii) [3 marks] Evaluate the following limit or show that the limit does not exist.

$$\lim_{(x,y)\to(0,0)} \frac{3xy - y}{x^2 - 3x + y^2 + 7y + 1}.$$

(iv) [4 marks] At the point defined by $t = a$ the tangent vector of the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} \cos(3t^2) \\ t \\ t^2 + 2t \end{pmatrix},$$

is a scalar multiple of the gradient vector $\nabla f(1, 2, 3)$ where $f(x, y, z) = x^2 + y^2 - 2x - 2yz$. Find the value of a.

(v) [4 marks] Find the directional derivative of the function

$$f(x, y) = \begin{cases} \frac{x^3y^2 - 3y^5}{x^4 + 2x^2y^2 + y^4}; & \text{if } x^2 + y^2 \neq 0 \\ 0; & \text{if } x^2 + y^2 = 0, \end{cases}$$

in the direction of the vector $\langle 1, -1 \rangle$ at the point $(0, 0)$.

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QUESTION 2. **(16 marks)**

(i) [4 marks] Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y)\to(0,0)} \frac{x^3y}{x^3y+(x-y)^4}$$

(b)

$$\lim_{(x,y)\to(1,0)} \frac{7x^2-7}{x^2+y^2-1}$$

(ii) [4 marks]

(a) Find k such that the function

$$f(x,y) = \begin{cases} \frac{x^4+3x^2-xy^3+3y^2}{x^2+y^2} & \text{if } x^2+y^2>0; \\ k & \text{if } x^2+y^2=0 \end{cases}$$

is continuous. (No justification is needed).

(b) For the k found in Part (a), prove that

$$\lim_{(x,y)\to(0,0)} \frac{x^4+3x^2-xy^3+3y^2}{x^2+y^2} = k.$$

(iii) [8 marks] Define

$$f(x,y) = \begin{cases} \frac{x^3-xy^2}{x^2+y^2}; & \text{if } x^2+y^2\neq 0 \\ 0; & \text{if } x^2+y^2=0. \end{cases}$$

(a) Is f continuous at $(0,0)$? Justify your answer.

(b) Find $f_x(0,0)$ and $f_y(0,0)$.

(c) Is f differentiable at $(0,0)$? Justify your answer.

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QUESTION 3. **(17 marks)**

(i) [3 marks] Find an equation for the tangent plane to the surface given by the equation $z = \ln(8x^2 + 9y^2 + 1)$ at the point $(0, 0, 0)$.

(ii) [3 marks] Use the Chain Rule to find $\dfrac{\partial u}{\partial y}$ at $(x, y, z) = (3, 2, 0)$ for the function

$$u(p, q, r) = e^{pq} \cos(r),$$

where

$$p = \frac{1}{x}, \quad q = x^2 \ln y, \quad r = z.$$

(iii) [3 marks] Assume that z is uniquely determined by the equation

$$9 \sin(x + y) + 4 \sin(x + z) + 3 \sin(y + z) = 0.$$

Find the value of $\frac{\partial z}{\partial x}$ at the point $(2\pi, -\pi, 4\pi)$.

(iv) [3 marks] In which directions is the directional derivative of

$$f(x, y) = \sin(xy) - x^2 + xy$$

at $(\pi, 0)$ equal to zero? Write the unit vector for each direction.

(v) [5 marks] Let

$$f(x, y) = 2y^3 + x^2 - 6xy - 8x + 4.$$

(a) Find all critical points of $f(x, y)$.

(b) For each critical point found in Part (a), determine whether or not it is a local max/min or a saddle point.

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